Analyzing Mission Impacts of Cyber Actions (AMICA)

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ABSTRACT

This paper describes AMICA (Analyzing Mission Impacts of Cyber Actions), an integrated approach for understanding mission impacts of cyber attacks. AMICA combines process modeling, discrete-event simulation, graph-based dependency modeling, and dynamic visualizations. This is a novel convergence of two lines of research: process modeling/simulation and attack graphs. AMICA captures process flows for mission tasks as well as cyber attacker and defender tactics, techniques, and procedures (TTPs). Vulnerability dependency graphs map network attack paths, and mission-dependency graphs define the hierarchy of high-to-low-level mission requirements mapped to cyber assets. Through simulation of the resulting integrated model, we quantify impacts in terms of mission-based measures, for various mission and threat scenarios. Dynamic visualization of simulation runs provides deeper understanding of cyber warfare dynamics, for situational awareness in the context of simulated conflicts. We demonstrate our approach through a prototype tool that combines operational and systems views for rapid analysis.

1.0 INTRODUCTION

In the U.S. Department of Defense (DoD) roadmap for cyber modeling & simulation (M&S), planning for integrated cyber and kinetic mission assurance is a key capability area [1]. The range of capabilities called out in the roadmap underscores the urgent need for rapid progress in this area, especially given the asymmetric nature of cyber conflict.

Of particular importance is the integration of kinetic operations with the defensive cyber operations that support them. This requires effective communication of cyber situations (and their big-picture impacts) to decision makers. In addition, there are numerous potential applications of cyber M&S, along a spectrum of increased maturity and corresponding research challenges, as shown in Figure 1.

Understanding mission resilience to cyber warfare requires bringing together layers of information from numerous sources. At the lower layers, network topology, firewall policies, intrusion detection systems, system configurations, vulnerabilities, etc., all play a part. We can combine these into a higher-level attack graph model that shows transitive paths of vulnerability. We also need to map cyber assets to mission requirements, and capture dependencies from low-level requirements to higher-level ones appropriate for decision making. Because mission requirements are highly dynamic, we need to capture time-dependent models of mission flow. Cyber attacks and defenses are similarly dynamic, and defenses generally vary depending on particular attack classes.

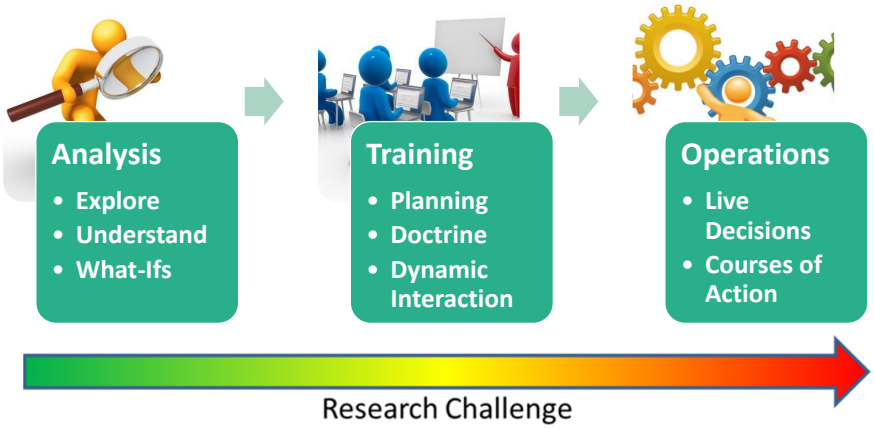

Figure 1: Spectrum of cyber M&S applications and challenges.

We introduce an approach that addresses all these aspects of mission-oriented cyber resilience, through an integrated M&S environment. This approach is called *Analyzing Mission Impacts of Cyber Actions* (AMICA). AMICA supports exploration and experimentation of the mission impacts of cyber warfare. The goal is to develop a flexible, extensible, modular, multi-layer M&S system for quantitative assessment of operational impacts of cyber attacks on mission performance. AMICA is expected to increase our understanding of dependencies between operational missions, cyber TTPs, and computing infrastructure.

2.0 PREVIOUS WORK

There have been numerous information-centric military exercises with aspects of mission assurance and cyber warfare. In many exercises (e.g., Global Thunder [2] and Turbo Challenge [3]), cyber security is an important component, but not the primary exercise focus. More cyber-focused exercises such as Cyber Flag [4] have integrated cyber activities with operational missions for training purposes.

M&S has been applied in more traditional military spheres, e.g., for inferring enemy intent [5], entity-based battlefield simulations [6], and command decision support [7]. However, military mission planning has yet to leverage M&S and other formal methods as part of its standard practice, especially in the area of developing cyber defensive courses of action. In short, tools such as AMICA for assessing mission impact of cyber warfare are generally unavailable for operations-level support. The defense community is aggressively accelerating cyber defense forces [8], further motivating the need for more advanced capabilities in cyber course-of-action planning.

In the cyber domain, M&S capabilities are still relatively immature. Still, previous work can be leveraged for certain components of an integrated overall M&S approach. Systems such as Topological Vulnerability Analysis (TVA) [9][10], Network Security Planning Architecture (NetSPA) [11], and NRL's ACCEPT (A Configurable Cyber Event Prioritization Tool) [12] fuse network data (topology, firewall rules, asset inventories, vulnerability scans/databases, intrusion alerts, etc.) into graph-based models for mapping vulnerability paths and prioritizing events. Capabilities such as MITRE's Cyber Command System (CyCS) [13] and Cyber Mission Impact Assessment (CMIA) [14], and AFRL's Cyber Mission Assurance [15] capture mission and cyber dependencies.

Another key enabler for cyber M&S is standardization efforts. Making Security Measurable™ [16] is a collection of standardization activities within the cyber security community. It includes Common Vulnerabilities and Exposures (CVE), Common Attack Pattern Enumeration and Classification (CAPEC), Cyber Observable Expression (CybOX), Structured Threat Information Expression (STIX), and many others. These standards cover different aspects of security data needed for building comprehensive and accurate models.

To capture the flow of mission and cyber processes, we leverage the Object Management Group (OMG) Business Process Model Notation (BPMN) [17] standard. We employ the commercial tool iGrafx [18], which extends BPMN with behavioral modeling, critical-path analysis, discrete-event simulation, Monte Carlo analysis, and experiment design.

3.0 APPROACH

To explore the AMICA approach, we are conducting a pilot study and developing a proof-of-concept system. We seek a flexible, extensible, modular, and multi-layer M&S environment for quantitative assessment of operational impacts of cyber attacks on specific missions, as shown in Figure 2. Thus components can be interchanged, e.g., multiple missions on an infrastructure, to support analysis of different questions.

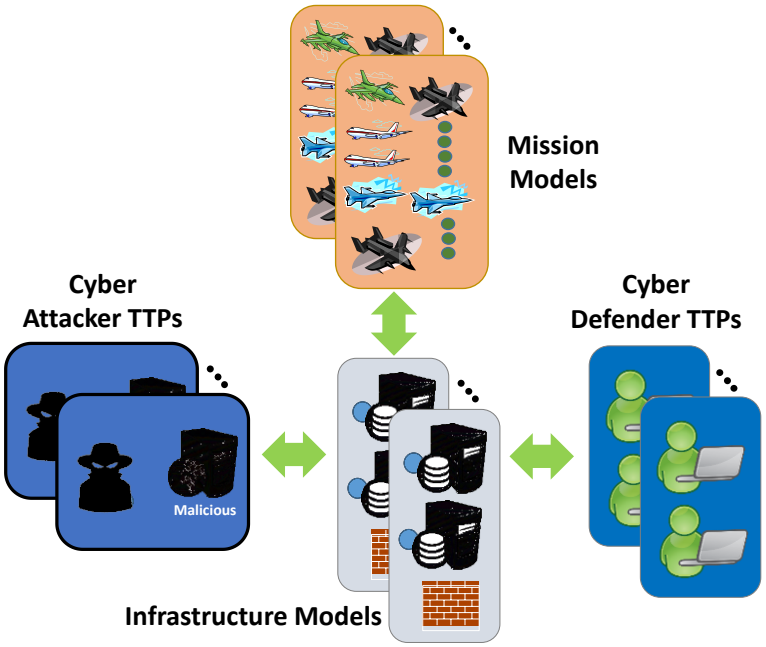

Figure 2: Modular libraries for model components.

AMICA currently includes libraries for operational (kinetic) missions, computing infrastructure on which missions depend, cyber attacker TTPs, and cyber defensive TTPs. Calibration and validation of the model occurs in concert with mission commanders, operators, and cyber defenders. In essence, we are connecting cyber effects to the kinetic domain, in the context of highly dynamic cyber warfare and mission threads. This helps commanders better maintain mission effectiveness in a force-on-force cyber-contested environment, and align defenses for best operational effect across a campaign.

For mission analysts and commanders, we seek to answer questions such as the following:

- When and where would be the most damaging attacks against the mission?
- How long before a particular attack has significant mission impact?
- How long does it take a mission to recover from an attack?
- What is more damaging to the mission: loss of reach-back availability or degradation of system assets?

For cyber defenders and analysts, we consider questions such as the following:

- What is the impact of better sensor performance, sensor location, etc.?
- How does a change to the network topology affect security posture?
- How well does the defense perform against different tiers of attacker?
- What is the impact of different defender TTPs?
- How to align workforce to cyber workload?
- What is the impact of adversary attack speed?
- What is the impact of adversary attack timing?

As illustrated in Figure 3, we employ a layered modeling structure. This allows inputs at both the operational and cyber layers to influence the behavior of the systems layer, to produce a combined effect on mission performance.

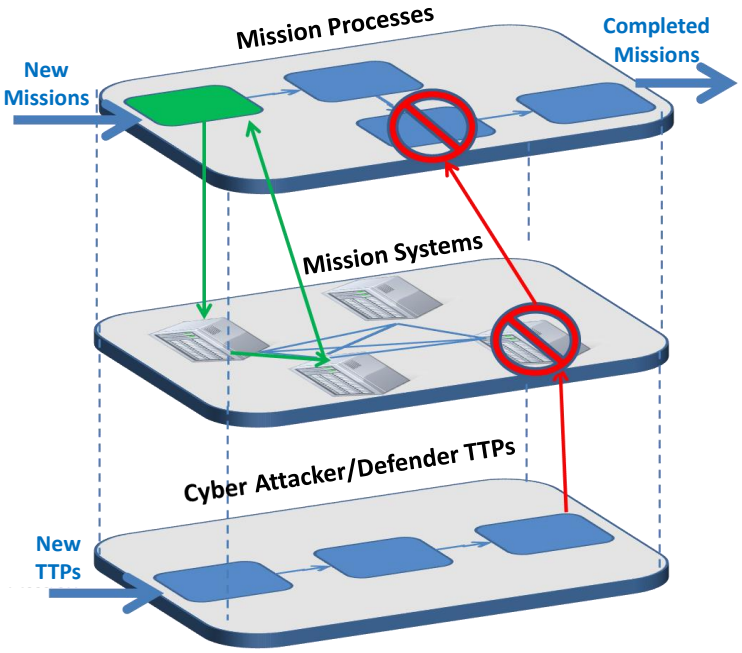

Figure 3: Modular libraries for model components.

Decoupling via layers provides model independence, with shared interfaces. This enables easy migration of missions and cyber TTPs as situations dynamically evolve. Figure 3 is notional only, and does not include all the model layers actually in AMICA. For example, there are layers for mission hierarchical dependencies, cyber vulnerability dependencies (attack graphs), etc.

Figure 4 shows the architectural structure of our AMICA implementation. This illustrates AMICA's novel approach for blending workflow modeling with mission dependencies and attack graphs. Each modality (process-based and graph-based) captures a different aspect of the overall picture: workflow (process modeling) and environment (graph-based relationships, constraints, etc.). This allows workflow and environment models to be developed independently, aided by automatic generation for a given network.

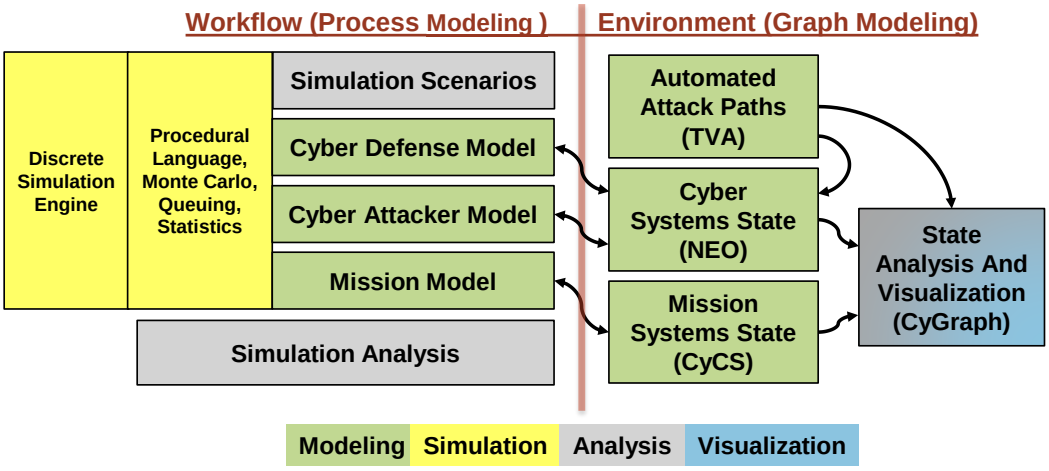

Figure 4: AMICA architecture.

Behavioral and temporal aspects of the system (workflow, timing constraints, required resources, etc.) are implemented through executable process models and stochastic discrete-event simulation (in iGrafx). Structural and functional aspects (environmental constraints, mission and system dependencies, event flows, etc.) are maintained through MIT Lincoln Laboratory's Network Environment Oracle (NEO), and MITRE's Cyber Command System (CyCS) [13] and CyGraph [19]. CyCS contains a directed graph comprising the information and system dependencies of each mission function. NEO contains additional topological and vulnerability information that is not captured in CyCS. CyGraph provides topological and attack graph-focused visualization of the environment and cyber attack progress. The initial state of the structural cyber (attack graph) model is generated from the network topology, firewall rules, and system vulnerabilities via the Government Off-The-Shelf (GOTS) tool TVA [9][10]. In this way, we leverage established tools for dependency knowledge management and automated model building.

To capture workflows, decision points, workloads, resources, and temporal constraints, AMICA employs a technique called Mission-Level Modeling (MLM) [20]. MLM leverages BPMN to define, refine, and verify operational processes, decisions, and information flows among producer/consumer systems and people. It supports model libraries and parameterization to quickly assemble new prototypes. MLM handles the high degree of concurrency inherent in information-sharing operations, and explores impacts on MOEs/MOPs through simulation of mission models.

MLM is based on BPMN and discrete-event simulation, implemented in iGrafx (a commercial tool). MLM replaces static tools such as Visio and PowerPoint, providing an executable, visual model to support stakeholder collaboration to develop and validate new concepts. This provides a single model for qualitative and quantitative analysis, and enables rapid prototyping and reuse thorough a single modeling standard.

Figure 5 shows the operational flow among the AMICA sub-systems. The TVA tool [9][10] provides the network topology and vulnerable attack paths through the network. This represents the initial state of the network, before cyber attacks and defenses are simulated. TVA initializes NEO, which maintains dynamic cyber state under simulation and provides choices for next possible cyber states. Similarly, CyCS maintains dynamic simulation state for mission dependencies.

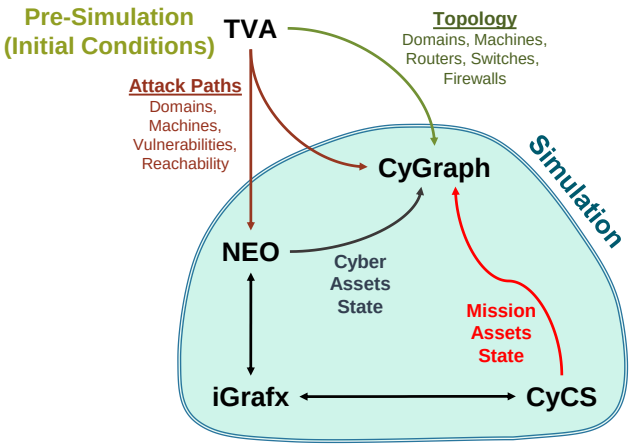

Figure 5: AMICA operational flow.

At simulation time, iGrafx simulates mission and cyber threads concurrently, testing cyber and mission states as needed, and updating them when process tasks (i.e., cyber attacker and defender tasks) change environmental conditions. For example, when the cyber attacker process compromises a mission-critical machine, iGrafx updates the node's state in CyCS (which propagates to higher-level mission dependencies).

Similarly, if the cyber defender process repairs the machine, its state is reset in CyCS. Asynchronously, mission tasks check the appropriate higher-level CyCS nodes upon which they depend. Throughout the entire process, CyGraph shows the dynamic state evolution through animated visualization.

4.0 CASE STUDY

For our case study, we consider a key mission within a regional Air and Space Operations Center (AOC). In an AOC, an air component commander provides top-level command and control of air and space operations. In our case study, the mission focus is deliberate kinetic targeting [21], from basic target development through development and publication of the Air Tasking Order (ATO).

Thus we model, simulate, and quantitatively analyze the impact of cyber attacks on the targeting mission (number of targets successfully processed) within an AOC. Our parameterized library of AMICA modules can be rapidly reconfigured to represent different mission, cyber threat, and/or cyber defense scenarios.

Figure 6 shows the phases of progression for target development. On-going target development defines all possible targets available for strike in the area of responsibility (AOR). In preparation for an anticipated crisis, advanced target development reexamines potential targets in preparation for possible strike.

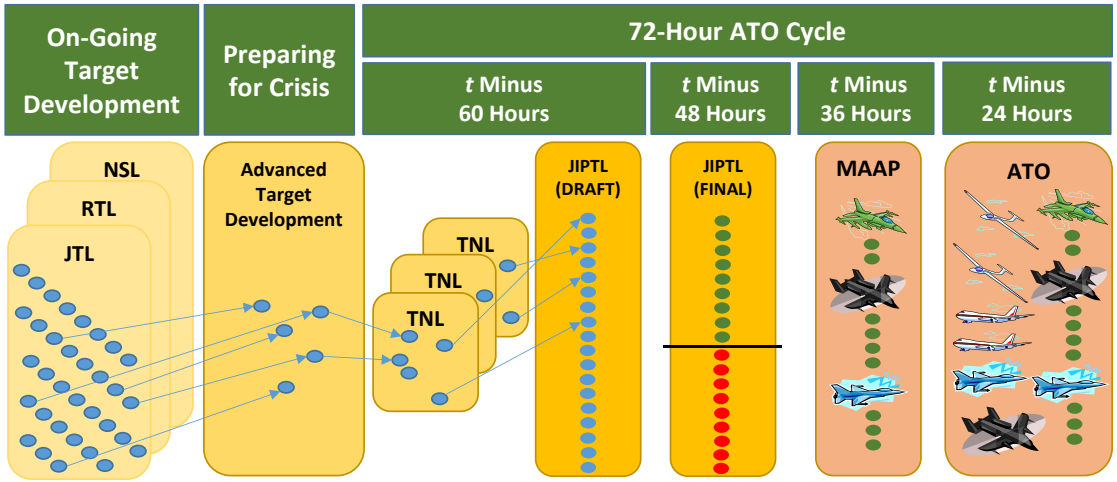

Figure 6: Target development and ATO process.

Once hostilities actually begin, targets are nominated for potential inclusion in the ATO. Nominated targets are prioritized, and then a final target is selected based on available delivery assets. Targets are paired with assets, leading to the completed ATO.

For this case study, we leverage Mission-Level Modeling (MLM) originally developed for U.S. European Command (EUCOM) for Exercise Austere Challenge 2010 [22]. This covers the targeting process from basic target development through the Master Air Attack Plan (MAAP) and ATO, as well as Battle Damage Assessment (BDA). This targeting model has over 200 steps, with timing and required resources per step. The model is organized as high-level modules that reference lower-level reusable library models. Figure 7 shows a high-level portion of this model.

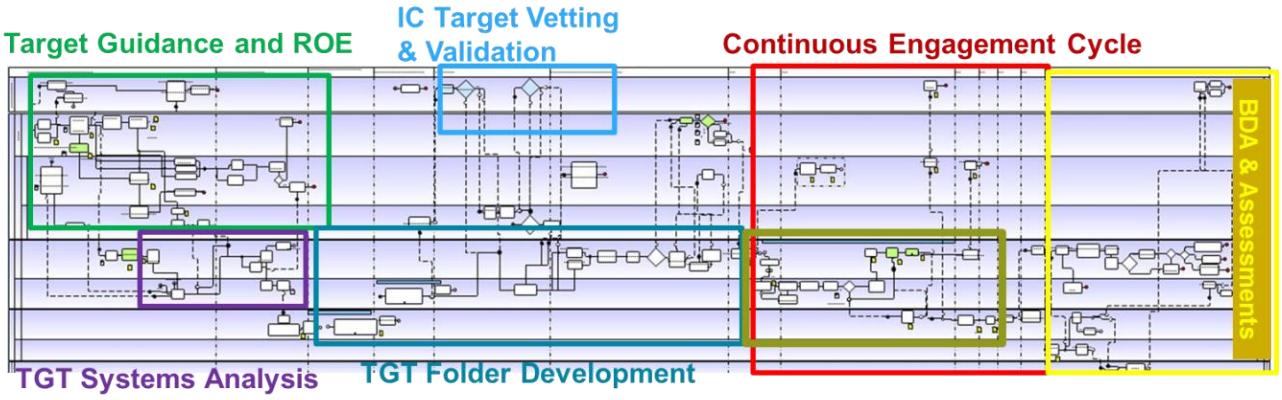

Figure 7: Portion of ATO target development model.

In this model, each target is tracked through the target-development process until completion, including whether the confidentiality or integrity of the target data was breached. Through simulation, we quantify mission performance and effectiveness, with metrics such as numbers of targets making each list, timing of each phase of development, workforce utilization, downtime, etc.

Figure 8 shows a high-level portion of a cyber attacker model. In this particular scenario, a phishing attack results in a malware infection, giving the adversary an initial presence inside the network. The attacker then moves laterally through the network, until a mission-critical machine is compromised. At that point, the attacker achieves the desired attack goal (compromising confidentiality, integrity, and/or availability). Depending on the scenario settings, the adversary may delay the final impact to coincide with a critical phase of the mission.

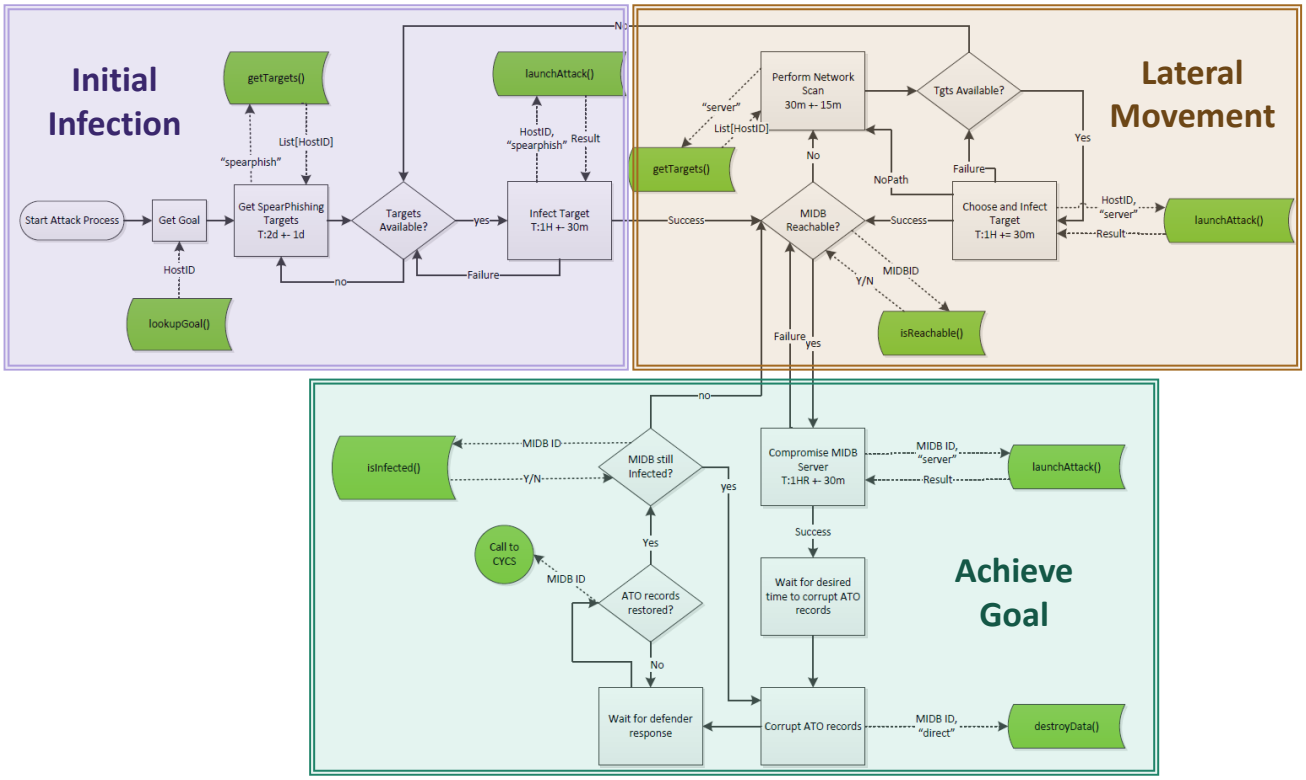

Figure 8: Portion of cyber attacker model.

Figure 9 shows a high-level portion of the cyber defender model. The process is triggered by an alert (intrusion detection system, user tipoff, etc.), followed by triage to understand the basic nature of the alert.

Depending on the severity of the incident and past history with the victim machine, the defender either reboots the machine, restores corrupted data, or rebuilds the machine from a non-compromised image. If an infection is detected or a machine is a victim in multiple incidents, the defender conducts more in-depth forensics. This involves searching for other infections and rebuilding victims as needed.

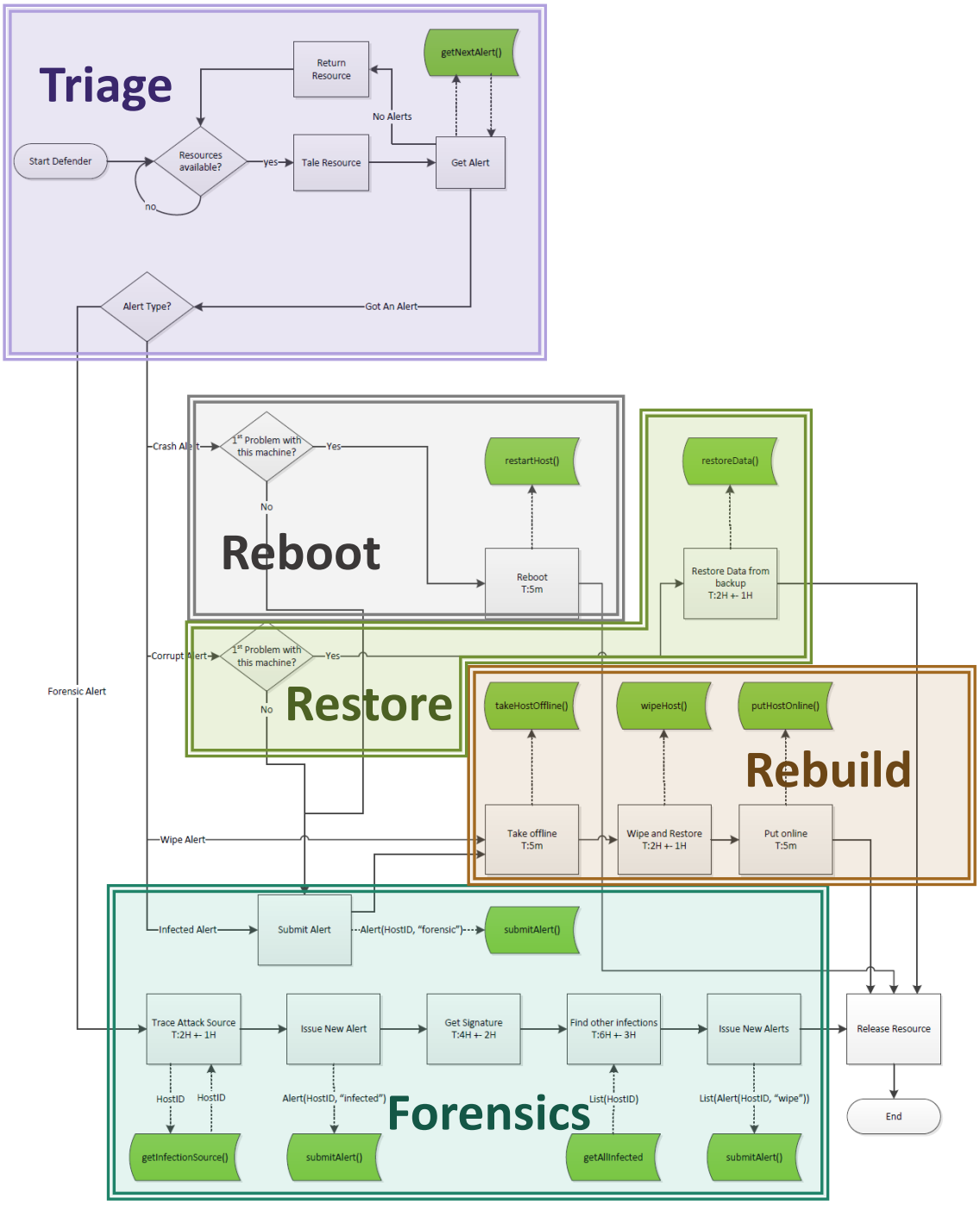

Figure 9: Portion of cyber defender model.

As for the mission model, the cyber (attacker and defender) models are modular, with higher-level models referencing sub-models. That is, process tasks (boxes) in a given model may represent entire sub-models defined elsewhere in the AMICA library.

Our cyber model leverages previous collaborative work with cyber defenders to define a process flow for their operations. This model captures adversary TTPs for major classes of attacks (email-based, browser-based, and host-based), with corresponding defensive TTPs. This collaborative work has produced a rich process diagram (in Visio), approaching 1000 steps. For AMICA, we use this as the basis for an executable model in iGrafx.

The cyber attacker and defender processes (in iGrafx) interact through the Network Environment Oracle (NEO). NEO maintains state in the cyber attack graph, which the attacker and defender process models check for environmental conditions required for taking next steps (vulnerabilities, reachability, infection state, etc.).

NEO state is reflected in CyGraph [19], a MITRE tool for cyber graph analytics, interactive visualization, and animation. Figure 10 shows a representative attack graph in CyGraph, with infected machines in red and rebuilt machines in green.

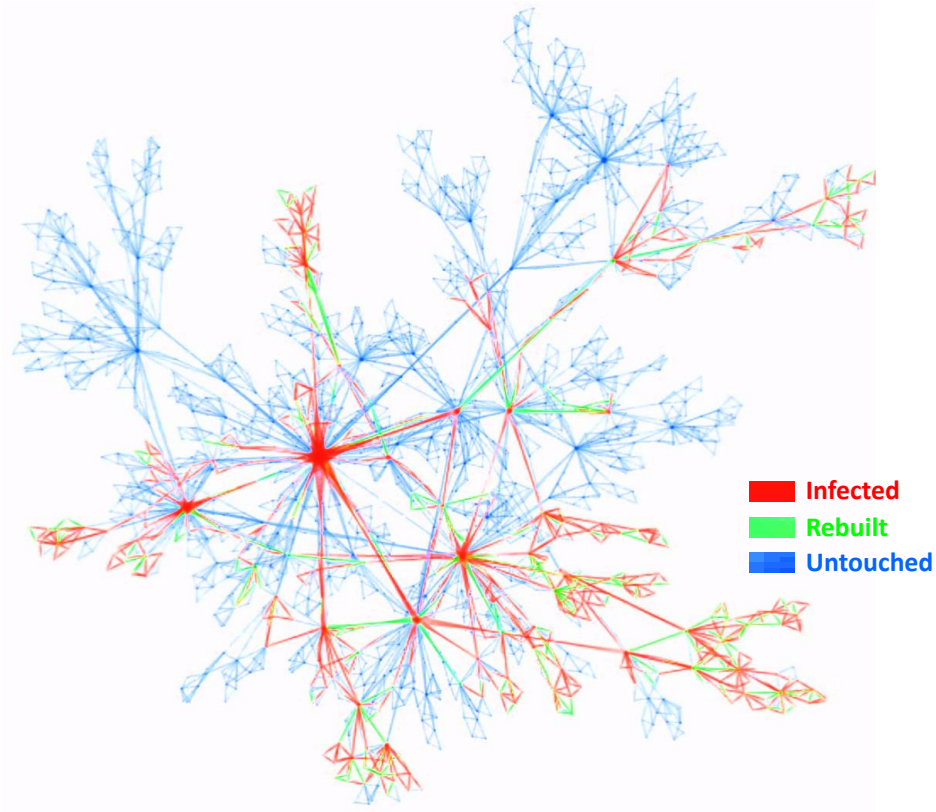

Figure 10: Cyber attack graph with dynamic states.

While NEO maintains state for cyber-related assets, MITRE's Cyber Command System (CyCS) maintains state for mission-related assets. CyCS models mission dependencies as a directed acyclic graph (hierarchy). The upper levels of the hierarchy are high-level mission assets (organizations, major work products, etc.). These are mapped to subordinate entities on which they depend. Dependencies can be conjunctive (Boolean AND) or disjunctive (Boolean OR). At the bottom of the hierarchy are those entities with no subordinates. Figure 11 shows a representative mission-dependency graph, visualized via CyGraph.

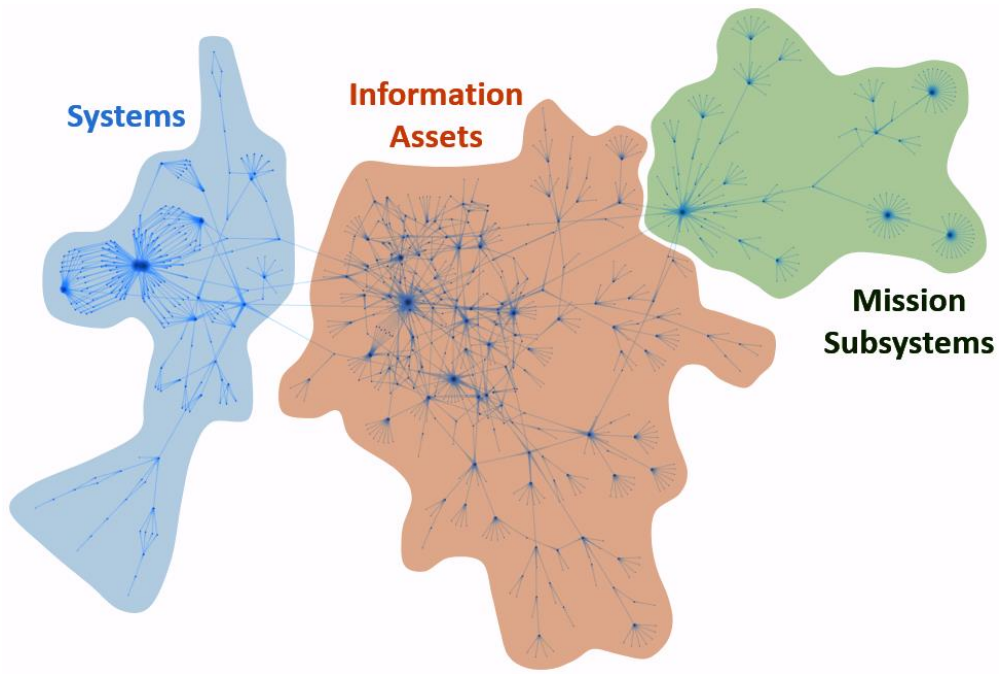

Figure 11: Graph of mission dependencies.

As an example of the quantitative analyses available through AMICA, consider Table 1. This shows mission impact from a simulated cyber attack. In this scenario, the attack results in loss of availability of a mission-critical database service.

Table 1-1: Impact of availability attack (JTL targets).

Cycle	Without Attacks	With Attacks	Relative Impact
4 days	9	1	88%
7 days	21	1	95%
14 days	76	70	8%

In this scenario, the attack occurs during routine operations early in the target-development process. The metric for cyber impact is a mission-based measure of performance (MOP): the number of targets that make the Joint Target List (JTL). The relative impact in the table (in percent) is then

$$Impact_{relative} = 100 \cdot [1 - (n_{with\ attacks}/n_{without\ attacks})].$$

The experiment is to determine a baseline number of JTL targets produced in the absence of an attack, and to compare that to the number of JTL targets produced when the AOC is under attack.

The results in Table 1-1 show a dramatic mission impact from the cyber attack. Moreover, the effects are fairly long-lasting; after a week, the relative impact is still only one JTL target produced (versus the expected 21 targets). By the end of the second week after attack, JTL target production is mostly caught up.

In these experiments, the processing of each target is simulated individually. At various points in the process, there are certain conditions, timings, etc., that have some degree of uncertainty. These are modeled as probability distributions in the appropriate points in the model. In a simulation run, Monte Carlo analysis executes the stochastic model according to model parameters.

Table 1-2 shows quantitative results from another AMICA simulation. This scenario is an integrity attack against a critical database during advance target development (the phase that prepares for an anticipated crisis). The mission-based MOP for measuring cyber impact is the number of targets added to the Joint Integrated Prioritized Target List (JIPTL).

Table 1-2: Impact of integrity attack (JIPTL Targets).

Cycle	Without Attacks	With Attacks	Relative Impact
4 days	574	303	47%
7 days	1098	1044	5%
14 days	1098	1087	1%

The results in Table 1-2 show that this attack is less impactful in terms of relative reduction in targets processed. Moreover, the AOC is able to rebound from the attack more quickly.

Figure 12 shows the relative impact on mission performance for the two attack scenarios: (1) availability attack against producing the JTL in routine early development, and (2) integrity attack against producing JIPTL in advanced target development in preparation for crisis.

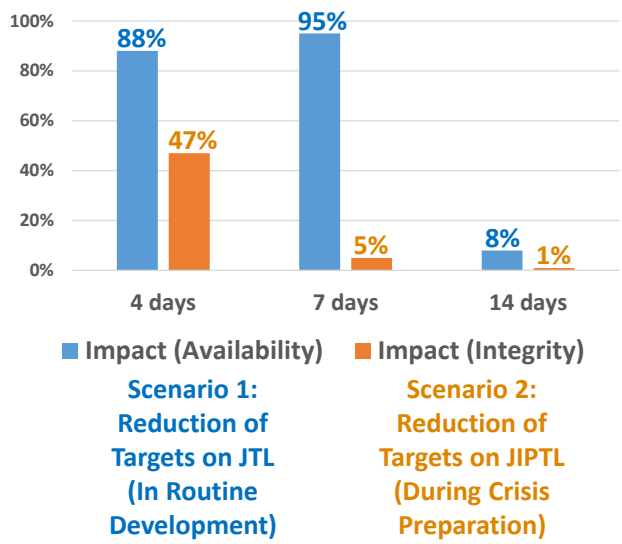

Figure 12: Relative impact for two attack scenarios.

Of course, not all target-production numbers may be equally important. For example, the criticality of the development phase itself may be a strong factor in overall impact. But it is clear that AMICA provides a

quantitative approach to address these kinds of questions, based on simulation of vetted models for missions and cyber TTPs.

We are investigating a range of more advanced attacks against different portions of the targeting process, such as data alterations that interfere with battle damage assessment, move target locations, inject discrepancies that force massive rework, etc.

5.0 SUMMARY AND NEXT STEPS

We have described an integrated approach for quantitative analysis of mission impact from cyber attacks, known as AMICA (Analyzing Mission Impacts of Cyber Actions). AMICA defines process models for mission threads and cyber tactics, techniques, and procedures (TTPs). These process models are designed as a hierarchically-decomposed library of reusable modules, for rapid reconfiguration and prototyping.

AMICA process models are probabilistic and executable, supported by discrete-event simulation and stochastic Monte Carlo analysis. Through simulation of mission and cyber models, we are able to quantitatively assess mission impact from cyber attacks. Monte Carlo analysis provides distributions over multiple simulation runs, for bounding uncertainty in results. For process modeling and simulation we apply industry-standard Business Process Modeling Notation (BPMN) implemented in a commercial tool (iGrafx).

While process models capture workflow and behavioral phenomena, processes necessarily operate within the structural constraints and dependencies of a particular environment. This includes dependencies between mission requirements and cyber assets, as well as constraints on attacker freedom of movement. We capture these through graph models (mission-dependency graphs and attack graphs), which are dynamically updated under process-model simulation.

This novel merging of M&S modalities supports dynamic simulation while leveraging established tools for cyber/mission knowledge management and automatic model building (e.g., attack graphs). Through simulation of this integrated multi-modal model, AMICA quantifies cyber impacts in terms of mission-based measures, for desired mission and threat scenarios. We provide animated visualizations of simulation runs, showing environmental state changes during the interplay of cyber force-on-force warfare.

We demonstrate AMICA through a case study, showing cyber impacts against a particular kinetic mission: targeting for Air Tasking Order (ATO) development in an Air and Space Operations Center (AOC). We model, simulate, and quantify the impact of cyber attacks on the targeting mission. We show impact results for two attack scenarios (availability and confidentiality) against different phases of the target-development process. Our simulations quantify cyber impact in terms of mission-relevant measures (numbers of targets completed) over time.

In the future, we plan to develop a more rigorous experimental framework for posing hypotheses, designing experiments, and validating results. The goal is to provide a rich and agile environment for gaining scientific insights. Examples of such hypotheses include the following:

- *Levels of Fidelity*: Given a threat model, what is the right level of fidelity to predict mission impact with sufficient accuracy?
- *Threat Classes*: For a given set of threat classes, what level of coverage is sufficient to maintain mission readiness?

- *Attacker TTPs*: What is the right degree of automation to achieve a desired mission impact? How much knowledge is required for a desired impact?
- *Attack Dynamics*: When should the adversary attack to have the highest mission impact? Which attack mode (e.g. fast smash-and-grab or slow-and-stealthy) can cause greater mission impact? How many concurrent attacks can the mission withstand?
- *Defense TTPs*: Under what conditions are static defenses inadequate? What is the best combination of static, dynamic, and synergistic defenses?
- *Attack Surface and Resiliency*: What degree of diversity gives adequate protection against zero-day attacks? What is the right balance between diversity, redundancy, containment, and cost?

Overall, AMICA merges cyber and kinetic domains (mission threads, cyber TTPs, network environment, etc.) into a common M&S environment, with complementary modeling modalities (process-based and graph-based). This provides a strong foundation for answering these kinds of questions about mission impact of cyber attacks and defenses.

6.0 ACKNOWLEDGMENTS

We would like to acknowledge the assistance provided by members of United States European and Pacific Commands, and the 603rd and 613th Air & Space Operations Centers, as well as Scott Foote of the MITRE Corporation. This work was performed in support of Dr. Steven King from Information Systems & Cyber Technology, Office of the Assistant Secretary of Defense (Research & Engineering) under contract W56KGU-14-C-0010. CyGraph was developed under the MITRE Innovation Program (project number EPF-14-00341), with Vipin Swarup as Innovation Area Leader. MIT Lincoln Laboratory work was performed under Air Force Contract FA8721-05-C-0002. Opinions, interpretations, conclusions, and recommendations are those of the authors and are not necessarily endorsed by the United States Government.

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